

Sankalp Rajeev

Dearborn, MI | (480) 208-2139 | srajeev@umich.edu | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

University of Michigan – Dearborn

M.Sc., Artificial Intelligence & M.S.E., Robotics Engineering; GPA: 3.90/4.00

Sep 2024 – Dec 2026

Dearborn, MI

Arizona State University

B.S., Computer Science; Minor: Mathematical Concepts of Engineering; GPA: 3.71/4.00

Aug 2020 – May 2024

Tempe, AZ

WORK EXPERIENCE

Magna International

May 2026 – Aug 2026

Data Science Intern, Advanced Manufacturing Innovation R&D

Troy, MI

- Validated a **multi-camera stamped-part PoC** with up to **\$7M/year** in projected labor savings at full production scale, proving feature registration and cross-camera deduplication on **2- and 4-camera rigs** (~20% overlap) and establishing the path to 6- and 24-camera systems using **SIFT/RANSAC** calibration and geometric/Hungarian matching.
- Delivered a **lens-debris inspection pipeline** projected to save **~\$700K/year** in labor, achieving **100% recall and 96% precision** with high-pass filtering + CLAHE preprocessing and PatchCore; handed the validated system off to the manufacturing plant.
- Delivered an **unsupervised visual-inspection pipeline** to a Magna Seating plant in Spain, achieving **100% defect recall, 95.2% precision, and ~300 ms end-to-end latency** on NVIDIA Jetson AGX Orin after benchmarking 6+ approaches and productionizing **YOLOX + NanoSAM + PatchCore** via ONNX, TensorRT FP16, Docker, and NVIDIA Triton.

Zoetis

May 2025 – Nov 2025

AI Engineering Intern

Kalamazoo, MI

- Deployed **4 production LLM tools** on Azure Kubernetes using asynchronous ETL and semantic matching, reducing regulated documentation effort by **1,000+ engineer-hours annually**.
- Reduced incident misclassification by **~40%** by building a real-time SQL-backed analytics workflow with Plotly dashboards and automated openpyxl Excel exports.

Mistral Solutions Pvt Ltd

Jun 2024 – Aug 2024

AI/ML Engineer Intern

Bangalore, India

- Reduced UAV inference latency by **25%** by quantizing YOLOv8 and ByteTrack with the Qualcomm SNPE SDK on a QRB5165 edge platform; built and deployed a **VGG16-based vehicle classification model (96% accuracy)** onto the same board via the SNPE pipeline.
- Enabled real-time range and thermal telemetry by integrating Lightware LW20/C LiDAR and DS1820 sensors through Arduino and I2C-based hardware-software interfaces.

TECHNICAL PROJECTS

Kairos – Demand Intelligence Platform | PySpark, Kafka, Airflow, PyTorch, XGBoost, Docker

- Processed **411M+ e-commerce events** via PySpark/Kafka/Airflow; built SASRec to Two-Tower to XGBoost recommendations (NDCG@10 ~**0.90**, AUC **0.9883**) and demand forecasting (**9.44% WMAPE**), validated with **GitHub Actions CI + 83 automated Pytest tests**.

Nexus – LLMOps Observability & Self-Healing Platform | LangGraph, Kafka, FastAPI, MLflow, Docker

- Built an **11-service agent reliability platform** that streams traces through Kafka, classifies **hallucination, context-bleed, and plan-drift** failures, performs LangGraph-based root-cause diagnosis and prompt/configuration repair, and re-evaluates fixes with DeepEval + MLflow in **<90 s**.

CodeMind – Multi-Agent Code Intelligence Platform | CrewAI, LangChain, ChromaDB, Ollama, QLoRA

- Built hybrid vector + BM25 + Reciprocal Rank Fusion retrieval achieving **15.4 ms median latency across 42K+ chunks from 10 repositories**, orchestrating specialized agents for code exploration, security, testing, performance, and documentation.
- Fine-tuned **Gemma 3 12B** with QLoRA/Unsloth for automated test generation, achieving **34.7% CodeBLEU (+73.5% relative vs. baseline)** with a **3.3x inference speedup**.

MeetingMind – Multimodal Meeting Intelligence | Vertex AI, LangChain, ChromaDB, Cloud Run

- Built a multimodal meeting pipeline with **Vertex AI Gemini 2.0 Flash**; cut VLM API cost by **85%** using an SSIM frame pre-filter and added RAG Q&A with **LangChain + ChromaDB + Vertex AI Embeddings**, deployed on Google Cloud Run via Artifact Registry.

TECHNICAL SKILLS

Computer Vision

Anomaly Detection, PatchCore, YOLOX, SAM2/NanoSAM, OpenCV, Object Detection

ML & Statistics

Machine Learning, Deep Learning, PyTorch, Transformers, XGBoost, scikit-learn, Unsupervised Learning

Edge & MLOps

NVIDIA Triton, Jetson AGX Orin, TensorRT, ONNX, OpenVINO, ClearML, MLflow

Data & Cloud

PySpark, Kafka, Airflow, Databricks, GCP (Vertex AI, Cloud Run, Artifact Registry), Azure Kubernetes Service, PostgreSQL, Redis

LLMs & Backend

LangGraph, LangChain, RAG, LLM Evaluation, Hugging Face, FastAPI

Languages

Python, C/C++, SQL, TypeScript, Java

DevOps/Tools

Git, GitHub Actions, CI/CD, Docker, Pytest, Linux